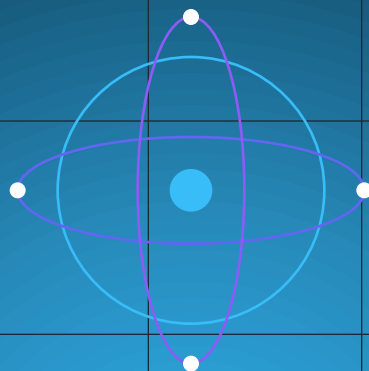




AMD RYZEN AI ACT II SPECIAL EDITION

NIMBUS AI

The Video Captioning Cloud that Never Leaves Your Browser

SUBMISSION TRACK 2: VIDEO CAPTIONING | BEST USE OF GEMMA IN VIDEO CAPTIONING

Traditional Cloud Pipelines

- Massive API Expenses: Enterprise-scale model endpoints impose major recurring hosting and server costs.
- Heavy Network Bottlenecks: Transmitting raw, high-resolution video files over public lines degrades bandwidth and adds latency.
- Compliance Risks: Shipping private, proprietary media assets to third-party servers presents fundamental security concerns.

The On-Device Edge Opportunity

- Zero Running Overhead: Computing entirely on-device eliminates ongoing subscription or server maintenance budgets.
- Absolute Data Privacy: Raw pixels never exit the client's web browser, establishing flawless offline compliance.
- Hardware Maximization: Harnesses local processing power of modern integrated AMD GPUs and Ryzen APUs.

Nimbus AI is an ultra-premium, 100% serverless local video video captioning cockpit engineered specifically to demonstration the edge-computing capabilities of Gemma 4 E2B IT.

By running completely within browser sandboxes utilizing WebGPU shader pipelines, Nimbus eliminates cloud communication overhead entirely. The cockpit provides real-time, rich multi-style copywriting with an interactive telemetry console.

\$0.00

CLOUD RUNNING COST

100%

CLIENT PRIVATE SECURITY

Our Technical Pillars

- Local Multimodal Core: Compiles custom shader pipelines dynamically to run Gemma 4 E2B IT.
- NVMe Caching: Sandboxes model parameters inside the Origin Private File System for immediate loading.
- Web Worker Execution: Decouples model inference onto background threads to allow responsive 120 FPS UI frames.

PRIMARY CHALLENGE : MODEL PIPELINE SERIALIZATION BUG

In the standard Transformers.js high-level pipeline('image-to-text') API, the preprocessor is unable to serialize raw image tensors and text prompts together for Gemma architectures, throwing empty inputs or shape exceptions.

Our Solution: We bypassed high-level wrappers entirely, writing direct lower-level bindings to coordinate model inputs.

01 / SEEK & EXTRACT

HTML5 Canvas programmatically captures balanced frames from the video timeline.

02 / AUTOPROCESSOR

Reads raw pixel matrices and compiles them directly into normalized tensors.

03 / INPUT INJECTION

Injects matched image placeholder objects manually into the message array.

04 / DIRECT GENERATE

Feeds multi-modal tensors directly to Gemma 4 E2B IT .generate() method.

AMD Power Governor

Active hardware throttling that scales image dimensions programmatically to regulate thermal footprints and system overhead.

- Turbo Mode: 384px resolution downscale for maximum visual clarity.
- Eco Mode: 224px downscale to restrict VRAM on AMD Ryzen APUs.

NVMe OPFS Caching

Bypasses slow browser storage limits to save weight variables inside Origin Private File System sandboxes.

- Direct Access: Sets use_opfs = true inside web environment.
- Subsequent Boots: Instant model boots using local NVMe read bandwidth.

Web Worker Offloading

Completely separates model computation from main render process to prevent UI lockups on heavy loads.

- Background Threads: Shaders execute in background threads.
- Fluid Render UI: Restores screen frame rates to fluid 120 FPS.

STAGE 1 / FACTUAL GROUNDING

Visual Tensor Analysis

Canvas programmatically samples the video timeline at temporal divisions. Images are converted into multi-modal tensors and passed to Gemma 4 E2B IT to generate a rich, chronological representation of the video scene.

- Detects physical environments and subject characteristics.
- Logs chronological actions and lighting variations.

STAGE 2 / CREATIVE COPYWRITING

Zero-VRAM Text Synthesis

Once factual grounding concludes, heavy visual tensors are immediately discarded to liberate WebGPU VRAM memory. Utilizing the lightweight text-only model pipeline, the system sequentially compiles descriptions into 4 styles:

FORMAL

SARCASTIC

TECH

NON-TECH

A Pioneering Milestone in Client-Side Edge Computing

Nimbus AI sets a new standard for WebGPU-accelerated desktop workflows. It proves that multimodal visual-language intelligence can run securely, instantly, and at zero server cost on standard consumer hardware.

By utilizing standard browser shaders without complex system configurations, Nimbus delivers a ready-to-run edge VLM solution that the world has never seen before.

Track 2 Submission Checklist

- ✓ Best Use of Gemma: Executes Gemma 4 E2B IT completely on-device, successfully bypassing the broken high-level JS preprocessor code.
- ✓ On-Device Hardware Innovation: Harnesses consumer Ryzen APUs and integrated Radeon graphics directly over standard browser WebGPU layers.
- ✓ Standardized Grader Pipeline: Python cloud agent container compiled and pushed to registry as v1000 with region-routed Gemini-3.5 support.